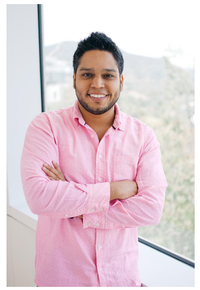
Google Developers

Experts

Data Analysis with BigQuery and Gemini



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Agenda

- Data Analysis
- Common Steps
- BigQuery and Gemini
- Demo

let's explain...

Data Analysis

- Data analysis is a systematic process of inspecting, cleaning, transforming, and modeling data to discover useful information, inform conclusions, and support decision-making

Data Analysis

Essential Steps in Data Analysis:

1. **Define the Problem:** Establish what you want to achieve or solve.
2. **Collect Data:** Obtain the necessary information.
3. **Clean Data:** Prepare the data by removing errors and inconsistencies.
4. **Transform Data:** Adapt the data for analysis, creating new variables if needed.
5. **Explore Data (EDA):** Understand patterns and characteristics with statistics and visualizations.
6. **Model Data:** Apply techniques to gain insights or make predictions.
7. **Interpret Results:** Make sense of the findings within the problem's context.

Bigquery Studio

BigQuery: Fast, Scalable Data Warehousing

- **Fully Managed & Serverless:** Google handles all infrastructure; you just focus on your data.
- **Massive Scalability:** Handles data from gigabytes to petabytes, scaling automatically.
- **Blazing-Fast Performance:** Runs complex SQL queries on huge datasets in seconds.
- **Cost-Effective:** Pay only for storage and queries, no fixed server costs.
- **Built-in ML:** Integrate machine learning directly with your data (BigQuery ML).
- **Secure & Integrated:** Robust security and seamless connection with other Google Cloud services.

BigQuery

Gemini

- **Multimodal:** Understands and processes text, code, audio, images, and video.
- **Highly Capable:** Google's most advanced and flexible model for diverse tasks.
- **Scalable Versions:** Available in sizes for various needs.
- **Enhanced Reasoning:** Excels at complex problem-solving and logic.
- **Broad Applications:** Opens new possibilities in research, content creation, and assistants.

DEMO

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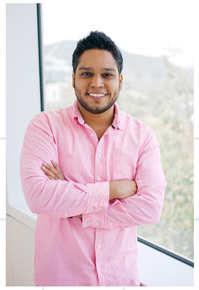
Thank You!
Gracias !



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